

Sodium-Ion Battery Chemistry: CATL Production & Emerging Technologies — A Referenced Investigation.

Matthias Roesslein — October 02, 2026

Executive findings

Premise correction — CATL's 2021 cathode was publicly identified as Prussian white, not "layered-oxide NMPA."

CATL's official July 29, 2021 announcement identified a redesigned **Prussian white** cathode and a porous **hard-carbon** anode for its first-generation sodium-ion cell. The supplied evidence does not establish an "NMPA" formulation for that product, and no such formulation should be reconstructed from generic layered-oxide research or patent families.[ref20; Tier 1 (High)] CATL subsequently described the newer Naxtra generation as using a **layered-oxide composite cathode** and long-cycle hard carbon, but the public evidence supplied here does not disclose a complete elemental formula, phase ratio, coating package, electrode recipe or bill of materials.[ref3; Tier 2 (Medium-High)][ref12; Tier 2 (Medium-High)]

This investigation uses information available through **2 October 2026**. Announced targets, contracts, certifications, prototypes, scheduled launches and projected production are kept separate from verified customer deliveries. In particular, a "mass-production vehicle" announcement or a supply contract does not by itself prove cumulative vehicle deliveries or installed energy-storage capacity.

The principal findings are:

1. CATL's sodium-ion program has at least three technically and commercially distinct stages.

The first-generation cell unveiled in July 2021 used Prussian white and hard carbon, reached up to 160 Wh/kg, charged to 80% state of charge in 15 minutes at room temperature, and retained more than 90% capacity at -20°C.[ref20; Tier 1 (High)] CATL also introduced the AB pack concept, in which sodium-ion and lithium-ion cells could coexist in one pack under a precision battery-management algorithm.[ref20; Tier 1 (High)] The 2021 statement set an industrial-chain objective rather than documenting a mature fleet deployment.

2. Naxtra is not merely a renamed first-generation cell. CATL introduced the Naxtra sodium-ion brand on April 21, 2025, claiming 175 Wh/kg for the passenger-vehicle cell, operation from -40°C to +70°C, 90% usable power at -40°C and more than 10,000 cycles.[ref11; Tier 1 (High)] Secondary technical reporting describes Naxtra as a layered-oxide-composite/hard-carbon system and identifies manufacturing challenges involving water control, hard-carbon gas generation and aluminum-foil adhesion.[ref12; Tier 2 (Medium-High)][ref222; Tier 2 (Medium-

High)] These descriptions should not be retroactively applied to CATL's explicitly Prussian-white 2021 cell.

3. **The production position at the cutoff is stronger than it was in 2021, but status language still matters.** CATL and Changan unveiled a Naxtra-equipped passenger vehicle on February 5, 2026 and described it as the first mass-production passenger vehicle using the technology, with market entry scheduled for mid-2026.[ref14; Tier 1 (High)] Reuters reported mass production and a planned fourth-quarter 2026 delivery ramp, while a 60 GWh, three-year agreement with HyperStrong was signed in April 2026 for stationary storage.[ref57; Tier 2 (Medium-High)][ref381; Tier 2 (Medium-High)][ref383; Tier 2 (Medium-High)] These sources establish industrialization, contracting and scheduled rollout; they do not provide an audited count of customer vehicles delivered or commissioned sodium-ion GWh by 2 October 2026.
4. **Shenxing and Shenxing Plus are lithium-ion products, not CATL sodium-ion chemistries.** CATL's official Shenxing Plus release describes an LFP battery exceeding 200 Wh/kg and supporting 4C fast charging.[ref65; Tier 1 (High)] The second-generation Shenxing line is likewise reported as LFP-based.[ref22; Tier 2 (Medium-High)] Naxtra is CATL's sodium-ion line, while Freevoy is an architecture capable of combining different electrochemical systems.[ref11; Tier 1 (High)] Any assertion that CATL integrated sodium chemistry into the 2024 Shenxing Plus cell is therefore unsupported by the supplied evidence. What was established in 2024 was an advanced LFP product; sodium/lithium integration traces to the 2021 AB concept and later Freevoy dual-power configurations, not to a sodium-ion Shenxing cathode.[ref20; Tier 1 (High)] [ref65; Tier 1 (High)]
5. **Freevoy and the AB pack are architectures, not single chemistries.** The AB concept places sodium-ion and lithium-ion cells in one managed pack.[ref20; Tier 1 (High)] Freevoy extends the multi-chemistry approach through independent energy zones and several possible combinations, including sodium-ion plus LFP, dual-LFP, and NCM-containing configurations. [ref11; Tier 1 (High)][ref59; Tier 3 (Medium)] A Freevoy pack must therefore be identified by its actual configuration before assigning it a cathode chemistry or energy density.
6. **CATL's complete sodium-ion recipe is not public in the evidence reviewed.** Public disclosures establish cathode families, hard carbon, use of aluminum current collectors on both electrodes, selected cell-level performance and broad manufacturing bottlenecks.[ref20; Tier 1 (High)][ref3; Tier 2 (Medium-High)][ref12; Tier 2 (Medium-High)] They do not establish the exact commercial NaPF₆ concentration, solvent ratios, additive package, separator polymer/coating, cathode stoichiometry, electrode loading, N/P ratio, porosity, formation protocol, cell volumetric energy density or complete mass-based bill of materials. Patent examples demonstrate technical possibilities, not necessarily the recipe in a shipping Naxtra cell.[ref356; Tier 1 (High)]
7. **The three principal sodium-ion cathode families occupy different design spaces.** Prussian blue analogues offer open three-dimensional diffusion frameworks, inexpensive synthesis and practical capacities commonly around 100–150 mAh/g, but water and vacancy defects can impair stability.[ref66; Tier 1 (High)][ref78; Tier 2 (Medium-High)] NASICON and related polyanionic materials trade lower baseline capacity for structural and voltage stability, while layered P2/O3 oxides offer high capacity but face phase transitions, air sensitivity and, in some

Mn-rich materials, Jahn–Teller distortion.[ref82; Tier 1 (High)][ref84; Tier 1 (High)][ref115; Tier 1 (High)]

8. **Hard carbon remains the leading commercial anode, but it is not a commodity-equivalent substitute for graphite.** Its storage response combines adsorption, insertion/intercalation and nanopore filling, while performance depends on pore structure, interlayer spacing, defects, surface area and solid–electrolyte–interphase formation.[ref394; Tier 1 (High)][ref399; Tier 1 (High)] Typical reported reversible capacities are approximately 250–350 mAh/g, but low initial Coulombic efficiency consumes cyclable sodium and may require cathode oversizing or pre-sodiation.[ref207; Tier 1 (High)][ref10; Tier 3 (Medium)]
9. **Generic electrolyte descriptions must not be presented as CATL formulations.** A representative research or modeled sodium-ion electrolyte is often about 1 M NaPF₆ in carbonate mixtures such as EC/DMC or EC/DEC, sometimes with FEC.[ref256; Tier 1 (High)] [ref258; Tier 1 (High)] Ether systems such as diglyme are also investigated.[ref253; Tier 1 (High)] The supplied public evidence does not demonstrate that CATL’s commercial Naxtra cells use any one of these exact formulations.
10. **The strongest near-term sodium-ion applications are those where cost exposure, low-temperature power, safety architecture or cycle throughput matters more than minimum mass and volume.** CATL’s Naxtra claims, HiNa’s urban vehicle deployment, and Prussian-blue-based high-power approaches illustrate distinct products rather than one universal sodium-ion performance profile.[ref11; Tier 1 (High)][ref123; Tier 3 (Medium)][ref201; Tier 2 (Medium-High)] Long-range vehicles remain more demanding because sodium-ion gravimetric and volumetric energy densities generally trail leading lithium-ion cells.[ref133; Tier 1 (High)][ref299; Tier 1 (High)]

Scope, method and reliability scheme

Scope and cutoff

The report examines CATL’s sodium-ion chronology, disclosed chemistry, manufacturing status, vehicle and storage integration, and the broader cathode, anode, electrolyte and cell-design landscape. It also profiles HiNa, SVOLT, AMTE Power, Northvolt, Faradion/Reliance and Natron, but only to the extent supported by the supplied evidence.

The factual cutoff is **2 October 2026**. Developments scheduled after that date are labeled as plans. A product launch is not treated as production; production is not treated as customer delivery; a supply agreement is not treated as installed capacity; and a laboratory result is not treated as a commercial-cell specification.

Reliability scheme

Tier	Definition used in this report	Treatment
Tier 1 (High)	Peer-reviewed journals; official company announcements and releases; patents and	Preferred for chemistry, official product claims and technical mechanisms. Company

Tier	Definition used in this report	Treatment
	official filings; authoritative intergovernmental publications	performance claims remain self-reported unless independently validated.
Tier 2 (Medium-High)	Reuters, C&EN, established battery and energy-industry publications	Used for independent commercial context, contracts and industrialization reporting.
Tier 3 (Medium)	Established technology and vehicle media, Wikipedia and verified LinkedIn material	Used for vehicle specifications and contextual leads when stronger evidence is absent; not used alone to establish sensitive chemistry conclusions.
Tier 4 (Low)	Blogs, vendor pages, forums and unverified commentary	Used only, if at all, to expose uncertainty or identify claims requiring verification.

An official company release receives Tier 1 because it is authoritative as to what the company announced, not because every performance claim has been independently reproduced. Peer-reviewed coin-cell results likewise receive Tier 1 for the experiment reported, but they do not automatically predict commercial large-format performance.

CATL chronology, current production and vehicle integration

2021: first-generation sodium-ion and the AB concept

On July 29, 2021, CATL unveiled its first-generation sodium-ion battery and AB battery-pack solution.[ref20; Tier 1 (High)] CATL explained that the larger sodium ion created material-structure and kinetic challenges and stated that it had redesigned the bulk structure of a Prussian white cathode by rearranging electrons to address rapid capacity fading.[ref20; Tier 1 (High)] The negative electrode was a proprietary porous hard carbon intended to provide abundant sodium storage and rapid movement.[ref20; Tier 1 (High)]

The announced cell reached up to 160 Wh/kg, could charge to 80% SOC in 15 minutes at room temperature, retained more than 90% capacity at -20°C and exceeded applicable national traction-battery thermal-stability requirements.[ref20; Tier 1 (High)] CATL also stated that the production process was compatible with lithium-ion manufacturing equipment and processes, enabling production lines to switch between the two systems.[ref20; Tier 1 (High)]

The AB battery pack was intended to mix sodium-ion and lithium-ion cells in one pack. A precision BMS algorithm would manage the two cell populations so that lithium-ion cells offset sodium-ion's energy-density limitation, while sodium-ion cells contributed low-temperature and power performance.[ref20; Tier 1 (High)] This was a pack-level integration concept. The announcement did not disclose a production vehicle, validated fleet volume, final cell ratio or complete pack architecture.

CATL said it intended to establish a basic sodium-ion industrial chain by 2023.[ref1; Tier 3 (Medium)] That statement was a target. It should not be converted into proof that a named vehicle or a specified number of cells had entered customer service by 2023.

2024: what was, and was not, integrated

CATL's Shenxing Plus announcement in 2024 concerned a **lithium iron phosphate battery**. CATL stated that Shenxing Plus crossed the 200 Wh/kg threshold for LFP, used granular gradation technology in the cathode and a three-dimensional honeycomb material in the anode, supported 4C charging and was designed for more than 1,000 km of range under the cited vehicle test basis. [ref65; Tier 1 (High)]

No supplied authoritative evidence identifies Shenxing Plus as sodium-ion, as an "NMPA" sodium cathode, or as a sodium/lithium mixed cell. The defensible conclusion is that CATL had:

- a sodium-ion program first presented in 2021;
- an AB mixed-cell pack concept from that launch;
- an LFP-based Shenxing Plus product announced in 2024; and
- later Naxtra and Freevoy offerings that formalized its multi-chemistry strategy.[ref20; Tier 1 (High)][ref65; Tier 1 (High)][ref11; Tier 1 (High)]

Thus, references to "2024 integration" require qualification. The evidence establishes parallel portfolio development in 2024, not incorporation of sodium-ion active material into Shenxing Plus.

2025: Naxtra launch and certification

CATL unveiled Naxtra on April 21, 2025 at its inaugural Super Tech Day.[ref11; Tier 1 (High)] It presented Naxtra as a mass-produced sodium-ion product family for passenger vehicles and heavy-duty commercial applications.[ref11; Tier 1 (High)] The passenger-vehicle cell was announced at up to 175 Wh/kg, which CATL described as comparable with LFP.[ref11; Tier 1 (High)]

CATL claimed operation from -40°C to +70°C, 90% usable power at -40°C, and no significant power degradation at 10% SOC.[ref11; Tier 1 (High)] It also claimed more than 10,000 cycles for the passenger product.[ref11; Tier 1 (High)] The 24 V integrated start-stop battery for heavy trucks was described as having a service life above eight years and reducing lifecycle cost by 61% relative to conventional lead-acid batteries.[ref11; Tier 1 (High)] Those figures are official product claims; the supplied evidence does not provide independent fleet-aging datasets reproducing them.

In September 2025, CATL announced that Naxtra had become the first sodium-ion battery to pass certification under China's GB 38031-2025 traction-battery safety standard.[ref47; Tier 1 (High)] Certification establishes compliance with the cited test regime; it does not disclose the complete internal formulation or prove field durability over 10,000 cycles.

2026: industrialization, Changan and stationary storage

CATL and Changan announced a Naxtra-equipped passenger vehicle on February 5, 2026.[ref14; Tier 1 (High)] CATL was identified as Changan's exclusive sodium-ion strategic partner across the

AVATR, Deepal, Qiyuan and UNI portfolios, and market availability was scheduled for mid-2026.[ref14; Tier 1 (High)] The companies described a 45 kWh configuration with more than 400 km of pure-electric range and presented 500–600 km as a future possibility as the supply chain matured.[ref14; Tier 1 (High)] The longer-range figures are projections, not specifications proven for the first configuration.

CATL reported that the battery retained more than 90% capacity at –40°C, could deliver stable power at –50°C and provided nearly three times the discharge power of an equivalent LFP battery at –30°C.[ref14; Tier 1 (High)] It also reported crush, drilling and sawing tests in which the battery continued supplying power without smoke or fire.[ref14; Tier 1 (High)] These are company-reported comparative and abuse-test results; exact protocols and complete third-party datasets are not included in the supplied evidence.

Secondary reporting associated the vehicle with the Changan Nevo A06.[ref24; Tier 3 (Medium)] Because the official CATL source supplied here establishes the Changan partnership but does not, in the summarized evidence, provide the same model-level detail, the Nevo A06 attribution should be treated as **reported but not independently verified here at the same level as the partnership itself**.

For stationary storage, CATL announced the TENER Sodium system in June 2026.[ref19; Tier 1 (High)] CATL described it as a field-validated sodium-ion energy-storage solution with modular storage durations and stated that a 60 GWh, three-year agreement with HyperStrong would support deployment.[ref19; Tier 1 (High)] Reuters independently reported the April 2026 supply agreement and a scheduled mass-delivery ramp in the fourth quarter of 2026.[ref381; Tier 2 (Medium-High)][ref383; Tier 2 (Medium-High)] At the cutoff, these sources support a major contracted pipeline and production ramp, not an assumption that all 60 GWh had been manufactured, delivered or commissioned.

Status ledger: announcement versus deployment

Item	Date	What is established	Verification level at 2 October 2026
First-generation CATL sodium-ion cell	July 29, 2021	Official unveiling; Prussian white/hard carbon; up to 160 Wh/kg; performance claims	Officially announced technology , not a documented named-vehicle deployment in the supplied evidence. [ref20; Tier 1 (High)]
AB battery pack	July 29, 2021	Official concept combining sodium-ion and lithium-ion cells under one BMS	Official architecture concept ; no supplied production-volume evidence. [ref20; Tier 1 (High)]
Shenxing Plus	2024	LFP battery above 200 Wh/kg with 4C charging	Official lithium-ion product ; not sodium-ion.[ref65; Tier 1 (High)]
Naxtra launch	April 21, 2025		

Item	Date	What is established	Verification level at 2 October 2026
		Sodium-ion product family; up to 175 Wh/kg; passenger and 24 V truck products	Official product launch and industrialization claim. [ref11; Tier 1 (High)]
GB 38031-2025 certification	September 2025	Naxtra passed the cited Chinese traction-battery standard	Official certification milestone. [ref47; Tier 1 (High)]
Changan Naxtra vehicle	February 5, 2026	Joint unveiling; market release scheduled for mid-2026	Official vehicle program; supplied evidence does not quantify customer deliveries by cutoff.[ref14; Tier 1 (High)]
HyperStrong agreement	April 2026	60 GWh over three years for energy storage	Binding commercial signal as reported, not delivered capacity. [ref381; Tier 2 (Medium-High)]
TENER Sodium	June 2026	Official stationary-storage product announcement	Official product, with rollout status to be assessed separately from announcement.[ref19; Tier 1 (High)]
Q4 2026 mass deliveries	Reported in 2026	Delivery ramp scheduled for the fourth quarter	Forward schedule at cutoff, not completed annual shipment evidence. [ref383; Tier 2 (Medium-High)]

CATL chemistry disclosure ledger

The central analytical discipline is to separate information disclosed for a named CATL product from generic sodium-ion design practice and from patent possibilities.

Component or metric	Explicit CATL disclosure or strongly attributed reporting	What remains undisclosed or should not be inferred
2021 cathode	Prussian white with redesigned bulk/electronic structure.[ref20; Tier 1 (High)]	Exact formula, vacancy concentration, crystal-water content, particle morphology, coating, electrode loading and supplier. No evidence for an “NMPA” formulation.
2021 anode	Porous hard carbon designed for sodium storage and rapid transport.[ref20; Tier 1 (High)]	Precursor, carbonization conditions, pore-size distribution, coating, binder, loading and pre-sodiation method.
Naxtra cathode	Secondary industry reporting identifies a layered-oxide composite.[ref3; Tier 2 (Medium-High)][ref12; Tier 2 (Medium-High)]	Exact Na/TM stoichiometry; O3/P2 fraction; Mn/Fe/Cu/Ni proportions; dopants, coatings and whether every Naxtra format uses the same cathode.
Naxtra anode		Exact carbon precursor, surface treatment, ICE, electrode formulation and whether “self-

Component or metric	Explicit CATL disclosure or strongly attributed reporting	What remains undisclosed or should not be inferred
	Long-cycle hard carbon is reported.[ref3; Tier 2 (Medium-High)]	forming anode” applies to every Naxtra product.
Current collectors	Aluminum on both electrodes is reported for CATL’s sodium-ion system.[ref3; Tier 2 (Medium-High)][ref12; Tier 2 (Medium-High)]	Foil alloy, thickness, surface treatment, tab design and mass share.
Electrolyte	CATL patents and reporting show broad work on sodium electrolytes and interface-forming additives.[ref356; Tier 1 (High)][ref134; Tier 3 (Medium)]	Commercial salt identity and concentration, solvent ratios, FEC or other additive content, water limit and electrolyte fill quantity.
Separator	CATL patents describe cells containing a separator and include possible porous-film structures.[ref356; Tier 1 (High)][ref373; Tier 1 (High)]	Naxtra separator polymer, thickness, porosity, shutdown behavior, ceramic coating and mass fraction.
Cell gravimetric energy	160 Wh/kg first generation; 175 Wh/kg Naxtra passenger cell.[ref20; Tier 1 (High)][ref11; Tier 1 (High)]	Test conditions, beginning-of-life basis, usable-versus-nominal basis and cell-format dependence are not fully reported in the supplied summaries.
Cell volumetric energy	No CATL-specific Naxtra value established in the supplied evidence	It must not be calculated from generic sodium-ion figures without cell dimensions, mass, void fraction and rated energy.
Cycle life	More than 10,000 cycles claimed for Naxtra passenger technology.[ref11; Tier 1 (High)]	Depth of discharge, C-rate, temperature, end-of-life threshold and rest protocol are not included in the supplied evidence.
Safety	CATL claims intrinsic safety and reports successful abuse testing and standards certification.[ref11; Tier 1 (High)][ref14; Tier 1 (High)][ref47; Tier 1 (High)]	Safety cannot be generalized to every sodium-ion cell independent of design, electrolyte, state of charge and pack propagation controls.
Complete BOM	Not publicly established	Cathode/anode masses, collector masses, electrolyte quantity, separator mass, casing, tabs and vent components are unavailable.

Patent disclosures should be interpreted carefully. A CATL patent establishing a positive electrode, negative electrode, separator and non-aqueous electrolyte proves that CATL sought protection for that architecture; it does not prove that the patent example is the production Naxtra recipe. [ref356; Tier 1 (High)] Likewise, patent references to amorphous carbon, aluminum-alloy current collectors or specialized separator laminates define options within an intellectual-property estate, not a confirmed cell teardown.[ref373; Tier 1 (High)][ref374; Tier 1 (High)]

Next-generation cathodes

Prussian blue analogues and Prussian white

Prussian blue analogues have open frameworks that support three-dimensional sodium-ion transport and high rate capability.[ref66; Tier 1 (High)][ref72; Tier 1 (High)] Prussian white is the sodium-rich, reduced form relevant to high initial sodium inventory. Reported theoretical capacity is approximately 170 mAh/g, while practical PBA capacities commonly fall around 100–150 mAh/g. [ref100; Tier 1 (High)][ref78; Tier 2 (Medium-High)] Representative nominal voltage for Prussian white is about 3.1–3.2 V.[ref99; Tier 1 (High)]

Their attractiveness comes from open diffusion channels, potentially low-cost aqueous synthesis and reliance on abundant elements in iron-based variants.[ref66; Tier 1 (High)][ref78; Tier 2 (Medium-High)] The same synthesis route creates key difficulties: $[\text{Fe}(\text{CN})_6]$ vacancies, coordinated or interstitial water, poor intrinsic electronic conductivity and structural changes during cycling can reduce practical capacity and voltage stability.[ref66; Tier 1 (High)][ref100; Tier 1 (High)]

Processing therefore matters as much as nominal composition. Controlled precipitation, vacuum drying, vacancy reduction, cation substitution, surface passivation and conductive composites have all been investigated to improve crystallinity, remove water and stabilize cycling.[ref99; Tier 1 (High)][ref103; Tier 1 (High)][ref105; Tier 1 (High)] For commercial due diligence, “Prussian white” is consequently insufficient as a performance specification: sodium content, vacancy fraction, water content, redox-metal selection, particle size and electrode density must be known.

CATL’s first-generation cell belongs in this family because CATL explicitly identified Prussian white. [ref20; Tier 1 (High)] The evidence does not establish that the same material remained the sole cathode in Naxtra. Secondary reporting instead describes Naxtra as a layered-oxide composite system.[ref3; Tier 2 (Medium-High)][ref12; Tier 2 (Medium-High)]

NASICON and other polyanionic cathodes

NASICON-type cathodes use a rigid three-dimensional framework that enables sodium transport while limiting structural collapse.[ref114; Tier 1 (High)][ref115; Tier 1 (High)] $\text{Na}_3\text{V}_2(\text{PO}_4)_3$, or NVP, is a common benchmark, with a theoretical capacity around 117 mAh/g and a voltage plateau near 3.4 V. [ref78; Tier 2 (Medium-High)] Fluorophosphate variants can raise operating voltage into approximately the 3.5–4.1 V range through the inductive effect of fluorine.[ref78; Tier 2 (Medium-High)]

The principal strengths are voltage stability, thermal and structural robustness, high-rate three-dimensional pathways and potentially very long cycle life.[ref115; Tier 1 (High)] The main weaknesses are lower baseline gravimetric capacity than the highest-capacity layered oxides, low electronic conductivity, and—in vanadium systems—material cost and toxicity concerns.[ref68; Tier 1 (High)][ref119; Tier 1 (High)] Carbon coating, particle-size reduction, active-element substitution and high-

entropy designs are used to raise conductivity, access additional redox couples and suppress lattice distortion.[ref115; Tier 1 (High)][ref119; Tier 1 (High)]

Advanced research variants have reported approximately 165 mAh/g, retention above 93% after 3,000 cycles at 20C in an Al-substituted framework, and other high-rate results extending to thousands of cycles.[ref115; Tier 1 (High)][ref119; Tier 1 (High)][ref120; Tier 1 (High)] These are material-specific laboratory demonstrations, not a generic performance guarantee for all NASICON commercial cells.

Iron-based phosphate/pyrophosphate systems seek to reduce vanadium dependence. Reported Fe-based variants show low volume change and long cycling, including results above 4,400 cycles.[ref78; Tier 2 (Medium-High)] Their likely value is in stationary and high-throughput applications where structural durability can justify lower specific capacity, but commercial conclusions require full-cell data at realistic loading and electrolyte quantity.

Layered O3 and P2 oxides

Layered sodium transition-metal oxides are denoted generally as Na_xTMO_2 and classified by oxygen stacking and sodium coordination.[ref84; Tier 1 (High)][ref106; Tier 1 (High)] O3 materials are comparatively sodium-rich and can offer high initial capacity, but sodium migration involves higher-barrier intermediate sites and cycling may induce O3-to-P3 or related transitions.[ref82; Tier 1 (High)][ref106; Tier 1 (High)] P2 materials commonly begin sodium-deficient and provide open prismatic migration sites, supporting faster diffusion and rate capability, but high-voltage desodiation can drive an irreversible P2-to-O2 transition.[ref84; Tier 1 (High)][ref85; Tier 1 (High)]

Manganese, iron, nickel and copper perform different functions:

- **Manganese** is abundant and can provide useful redox capacity, but Mn^{3+} can undergo Jahn-Teller distortion and contribute to structural instability.[ref84; Tier 1 (High)]
- **Iron** can reduce material cost and alter Na-O spacing and transition-metal-oxygen bonding, helping manage anionic redox and structural evolution in selected compositions.[ref88; Tier 1 (High)]
- **Nickel** can raise voltage and capacity but reintroduces a critical-material exposure that sodium-ion is sometimes advertised as avoiding.[ref133; Tier 1 (High)]
- **Copper** is used in selected layered compositions to tune redox response and stabilize multi-metal structures.[ref87; Tier 1 (High)]

A post-mortem study of a HiNa cell identified the layered composition $\text{NaCu}_{1/9}\text{Ni}_{2/9}\text{Fe}_{1/3}\text{Mn}_{1/3}\text{O}_2$. [ref91; Tier 1 (High)] This is important evidence that at least one commercial-oriented HiNa route used a Cu/Ni/Fe/Mn layered oxide. It does not establish that every HiNa product, or any CATL cell, uses that exact formula.

Reported P2 Fe/Mn materials can reach initial capacities as high as 243 mAh/g, but such values should not be treated as sustained commercial capacities without corresponding retention, voltage, loading and full-cell information.[ref78; Tier 2 (Medium-High)] Stabilization approaches include Ti or Mg substitution, high-entropy multi-cation design, surface coatings and P2/O3

intergrowths that combine O3 capacity with P2 kinetics.[ref82; Tier 1 (High)][ref87; Tier 1 (High)][ref107; Tier 1 (High)]

For Naxtra, “layered–oxide composite” is the highest defensible public chemistry resolution in the supplied evidence.[ref3; Tier 2 (Medium–High)] Assigning a specific O3/P2 structure or Mn/Fe/Cu/Ni ratio would be speculation.

Anodes and solid–state sodium systems

Hard carbon

Hard carbon is a non–graphitizable, disordered carbon consisting of turbostratic graphitic domains, defects and pores.[ref394; Tier 1 (High)][ref395; Tier 1 (High)] It is the leading sodium–ion anode because its expanded and disordered structure can accommodate sodium more effectively than conventional graphite.[ref158; Tier 1 (High)][ref361; Tier 1 (High)]

Its voltage profile is commonly divided into a higher–voltage slope and a low–voltage plateau. The slope is associated with adsorption at defects and insertion between disordered carbon layers; the plateau is strongly associated with sodium filling nanopores, although the detailed contribution of quasi–metallic clusters remains an active research question.[ref394; Tier 1 (High)][ref399; Tier 1 (High)]

Commercially relevant hard carbons commonly deliver approximately 250–350 mAh/g.[ref207; Tier 1 (High)] Advanced doped materials may report substantially higher laboratory values, including about 520 mAh/g, but these results should not be used as standard production–cell assumptions.[ref10; Tier 3 (Medium)] Hard carbon lacks a single practical theoretical–capacity number comparable to a crystalline one–electron insertion compound because capacity depends strongly on defects, interlayer storage and accessible pore volume.

The primary penalty is low initial Coulombic efficiency. Electrolyte reduction and SEI formation irreversibly consume sodium, while some sodium can become trapped in the carbon structure.[ref394; Tier 1 (High)][ref395; Tier 1 (High)] A full cell may therefore require extra sodium inventory through cathode oversizing, sacrificial salts or pre–sodiation, each of which changes its mass, manufacturing risk and safety profile.[ref269; Tier 1 (High)]

Higher surface area can improve kinetics but also increase electrolyte consumption. Closed–pore volume can raise plateau capacity, while pore–mouth dimensions influence desolvation and SEI chemistry.[ref399; Tier 1 (High)] Research has shown that suitably small pore mouths can encourage a thin, inorganic, NaF–rich interface and improve initial efficiency and cycling.[ref399; Tier 1 (High)]

CATL’s first–generation porous hard carbon is an explicit disclosure, but its pore architecture and precursor are proprietary.[ref20; Tier 1 (High)] Naxtra reporting also identifies hard carbon and manufacturing work to control gas generation.[ref12; Tier 2 (Medium–High)] That gas–generation

challenge reinforces why laboratory hard-carbon capacity alone cannot predict production yield or cell swelling.

Soft carbon and expanded graphite

Soft carbon is more ordered and graphitizable than hard carbon. It generally provides lower sodium-storage capacity but may have fewer reactive defects, lower electrolyte consumption and useful high-rate characteristics.[ref145; Tier 4 (Low)][ref154; Tier 1 (High)] Because part of that comparison is supplied through a low-tier source, it should be treated as directional rather than a procurement-grade benchmark.

Conventional graphite is poorly suited to sodium storage in ordinary carbonate systems because its interlayer spacing and sodium-graphite thermodynamics are unfavorable.[ref342; Tier 1 (High)] Expanded graphite increases layer spacing and is being investigated as a sodium host.[ref149; Tier 1 (High)][ref152; Tier 1 (High)] The supplied evidence does not show it displacing hard carbon in CATL's disclosed commercial system.

Alloy and conversion-type anodes

Tin, antimony, phosphorus and related alloying materials can offer much higher theoretical capacities than carbon.[ref149; Tier 1 (High)] Sodium-tin systems have been cited at a theoretical 847 mAh/g, compared with roughly 250–300 mAh/g for conventional hard carbon.[ref287; Tier 3 (Medium)] The drawback is extreme volume change: alloy anodes can expand by approximately 300–400%, causing pulverization, contact loss and rapid fading.[ref149; Tier 1 (High)]

Nanostructuring, carbon matrices and binary or ternary composites can buffer expansion.[ref149; Tier 1 (High)] These strategies add inactive mass and process complexity, so high active-material capacity does not automatically translate into higher full-cell energy or lower cost.

Titanium dioxide offers relatively high operating potential and is researched for safety and low-cost operation, but low ionic and electronic conductivity constrain performance.[ref146; Tier 1 (High)][ref151; Tier 1 (High)] Carbon coatings and conductive composites are used to improve kinetics.[ref146; Tier 1 (High)]

Self-forming and anode-free concepts

CATL's later disclosures reference "self-forming anode" technology, and reporting links it to higher gravimetric and volumetric energy density in Freevoy configurations.[ref59; Tier 3 (Medium)] Patent analyses also indicate CATL interest in anode-free designs in which sodium plates onto a current collector, potentially aided by carbon-nanotube or sodiophilic coatings.[ref134; Tier 3 (Medium)]

These concepts should not be conflated with the disclosed hard-carbon anode in every Naxtra cell. Anode-free sodium designs can reduce initial inactive mass, but uniform metal deposition, Coulombic efficiency, interface stability and dendrite control become central constraints. The supplied evidence does not establish that all commercial Naxtra passenger cells are anode-free.

Solid-state sodium batteries

Solid-state sodium systems replace the conventional liquid electrolyte and polymer separator with a solid sodium-ion conductor. Investigated electrolyte classes include sodium beta-alumina, NASICON ceramics, sulfides, polymers and composites.[ref185; Tier 1 (High)] Reported inorganic conductivities span approximately 10^{-4} to 10^{-2} S/cm, while stability windows can extend toward 5 V in selected systems.[ref185; Tier 1 (High)]

NASICON-type solid electrolytes such as $\text{Na}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$ are prominent because of their three-dimensional conduction pathways.[ref186; Tier 1 (High)] A reported $\text{Na}_5\text{GdSi}_4\text{O}_{12}$ material reached 1.9×10^{-3} S/cm at 30°C.[ref186; Tier 1 (High)] Such conductivity values characterize a material specimen; they do not establish full-cell power density.

The principal barriers are room-temperature conductivity, brittle ceramic mechanics, poor solid-solid contact and high electrode/electrolyte interfacial resistance.[ref185; Tier 1 (High)][ref186; Tier 1 (High)] Sodium-metal reactivity and volume change during plating and stripping can destabilize interfaces and create mechanical stress.[ref185; Tier 1 (High)] Buffer layers, surface treatments, composite electrolytes and in-situ polymerization seek to improve contact and suppress nonuniform deposition.[ref186; Tier 1 (High)]

No supplied CATL source establishes a commercially deployed all-solid-state sodium Naxtra cell by the cutoff. Solid-state sodium should therefore be treated as an emerging research path, not a feature of CATL's current disclosed sodium-ion products.

Electrolyte formulation

Representative liquid systems

The electrolyte must transport sodium ions while forming stable SEI and cathode-electrolyte-interphase layers.[ref360; Tier 1 (High)] A widely used research formulation is **1 M NaPF₆** in carbonate solvents. Examples include EC/DEC and other combinations containing EC, DMC, PC or EMC.[ref256; Tier 1 (High)][ref258; Tier 1 (High)] FEC is frequently studied as an interface-forming additive.[ref360; Tier 1 (High)]

Ether solvents such as diglyme can produce different solvation and SEI behavior and have shown improved initial efficiency or rate performance for selected electrodes.[ref253; Tier 1 (High)][ref254; Tier 1 (High)] Low-temperature research also examines PC-rich and multicomponent low-freezing-point systems because falling temperature increases viscosity, reduces ionic conductivity and raises desolvation and charge-transfer barriers.[ref306; Tier 1 (High)][ref308; Tier 1 (High)]

These formulations are **representative**, not universal. Cathode voltage, hard-carbon surface chemistry, separator wettability, aluminum stability, gas evolution and safety requirements determine the optimum salt, solvent and additive package.

What can be said about CATL

CATL has disclosed work on electrolyte and interface optimization, and its patents describe non-aqueous electrolytes and additive strategies.[ref356; Tier 1 (High)] Patent-landscape reporting also associates CATL with ether solvents and sodium borate salts.[ref134; Tier 3 (Medium)] However, the supplied evidence does not identify the commercial Naxtra electrolyte as 1 M NaPF₆, does not disclose solvent ratios and does not confirm a specific FEC concentration.

It is therefore appropriate to say:

- 1 M NaPF₆ in carbonate solvents is a common reference formulation in sodium-ion research.[ref256; Tier 1 (High)]
- Ether-based formulations are credible alternatives under active investigation.[ref253; Tier 1 (High)]
- CATL's exact Naxtra electrolyte remains proprietary or undisclosed in the evidence reviewed.

It is not appropriate to convert a patent example or laboratory convention into a Naxtra production BOM.

Separator systems

Polyethylene and polypropylene porous membranes are common sodium-ion separator candidates, while glass-fiber separators remain common in laboratory coin cells.[ref252; Tier 1 (High)][ref254; Tier 1 (High)] Porous PVDF and ceramic-functionalized separators are investigated to improve wettability, thermal stability and interfacial behavior.[ref267; Tier 1 (High)]

CATL patents confirm the architectural presence of a separator and discuss possible laminated porous films or heat-resistant layers.[ref356; Tier 1 (High)][ref373; Tier 1 (High)] They do not disclose the commercial Naxtra separator's polymer, ceramic loading, thickness or shutdown characteristics.

Current collectors

Sodium-ion cells can generally use aluminum foil for both positive and negative current collectors because aluminum does not undergo the problematic low-potential sodium alloying behavior that would prevent its use on the negative side.[ref258; Tier 1 (High)][ref261; Tier 1 (High)] Lithium-ion cells conventionally use aluminum at the cathode and copper at the graphite anode.

Using aluminum on both sodium-ion electrodes can reduce collector cost and mass, simplify deeply discharged shipping and avoid copper dissolution issues associated with overdischarged lithium-ion cells.[ref98; Tier 1 (High)][ref301; Tier 1 (High)] Zero-volt storage is not consequence-free: research on Prussian-white/hard-carbon cells found that prolonged 0 V storage can increase resistance and cause polarized capacity loss if the SEI is damaged, although reformation can recover performance.[ref98; Tier 1 (High)]

CATL reporting identifies aluminum collectors on both electrodes and highlights adhesion as one of the production challenges resolved during industrialization.[ref3; Tier 2 (Medium-High)][ref12; Tier 2 (Medium-High)] Exact foil gauge and surface treatment are not disclosed.

Cell-level material composition and mass fractions

Architecture

A conventional sodium-ion cell contains:

- cathode active material, conductive carbon and binder on a current collector;
- hard carbon or another anode material, conductive additive and binder on a current collector;
- a porous separator;
- sodium-salt electrolyte;
- tabs, enclosure, seals and safety hardware.[ref356; Tier 1 (High)]

Electrode production is broadly compatible with lithium-ion slurry mixing, coating, drying and assembly, which underpins the “drop-in” manufacturing argument.[ref20; Tier 1 (High)]

Compatibility does not mean identical process controls: Prussian-white water management, hard-carbon gas generation and dual-aluminum adhesion create sodium-specific constraints.[ref12; Tier 2 (Medium-High)]

Composition table and evidence limits

The supplied evidence does **not** provide an audited commercial sodium-ion whole-cell mass breakdown. The only defensible numerical formulation ranges concern electrode coatings in research examples. Those numbers must not be mistaken for whole-cell fractions.

Cell component	Representative chemistry-dependent material	Supported formulation or fraction	Correct basis and caveat
Cathode active material	Prussian white/PBA	Whole-cell fraction not supplied	Practical active-material capacity commonly 100–150 mAh/g; theoretical capacity about 170 mAh/g. [ref78; Tier 2 (Medium-High)][ref100; Tier 1 (High)]
Cathode active material	NVP/NVPF or related polyanion	Whole-cell fraction not supplied	Research NVP electrode examples use about 70–85 wt% active material within the cathode coating , not within the full cell.[ref263; Tier 1 (High)][ref264; Tier 1 (High)]
Cathode active material	Layered O3/P2 oxide	Whole-cell fraction not supplied	Fraction depends on formula, electrode density, sodium inventory

Cell component	Representative chemistry-dependent material	Supported formulation or fraction	Correct basis and caveat
			and N/P balance; no universal value is supported.
Cathode conductive additive	Super P, acetylene black, Ketjen black or related carbon	Approximately 10–15 wt% in cited research cathode coatings	Electrode-level range from selected NVP studies; commercial high-loading electrodes may differ.[ref263; Tier 1 (High)][ref264; Tier 1 (High)]
Cathode binder	PVDF or other binder	Approximately 5–15 wt% in cited research cathode coatings	Electrode-coating basis only; 15% is characteristic of some laboratory formulations and is not a presumed commercial value.[ref263; Tier 1 (High)][ref264; Tier 1 (High)]
Anode active material	Hard carbon	Whole-cell fraction not supplied	Typical reversible capacity approximately 250–350 mAh/g; initial sodium loss affects required mass balance.[ref207; Tier 1 (High)]
Anode binder	CMC/SBR, PAA or PVDF	Often a small electrode fraction; exact commercial range not established	Generic literature notes binder and conductive additive values often around 5% each in some formulations, but manufacturer recipes vary. [ref228; Tier 1 (High)]
Electrolyte	NaPF ₆ in carbonate solvents; ether alternatives	Whole-cell mass fraction not supplied	Fill quantity depends on porosity, separator and format; 1 M is a common concentration, not a CATL disclosure.[ref256; Tier 1 (High)] [ref258; Tier 1 (High)]
Separator	PE, PP, PVDF, coated membrane	Whole-cell mass fraction not supplied	Research coin cells often use glass fiber, which is not a reliable commercial-cell proxy.[ref252; Tier 1 (High)][ref254; Tier 1 (High)]
Positive current collector	Aluminum foil	Whole-cell mass fraction not supplied	Foil thickness and loading determine the fraction.
Negative current collector	Aluminum foil	Whole-cell mass fraction not supplied	A key sodium-ion distinction, but no CATL foil mass is public in the supplied evidence.[ref258; Tier 1 (High)]

Cell component	Representative chemistry-dependent material	Supported formulation or fraction	Correct basis and caveat
Enclosure, tabs and vent	Steel, aluminum or laminate depending on format	Whole-cell mass fraction not supplied	Cell format must be known before assigning a range.

A valid whole-cell mass model would require, at minimum, electrode area, coating loading, active-material ratio, current-collector thickness and density, separator dimensions, electrolyte fill, enclosure mass, tab mass and formation losses. None is comprehensively available for Naxtra in the supplied evidence.

Consequently, an analyst should not create a numerical CATL BOM by combining a laboratory NVP coating recipe with generic hard carbon and assumed pouch-cell hardware. That would mix incompatible chemistry and format bases.

Active-material specific capacities

Material	Theoretical or reference capacity	Demonstrated or practical capacity	Interpretation
Prussian white/PBA	Approximately 170 mAh/g theoretical.[ref100; Tier 1 (High)]	Commonly 100–150 mAh/g practical.[ref78; Tier 2 (Medium-High)]	Water, vacancies and conductivity cause the practical gap.
NVP	Approximately 117 mAh/g theoretical.[ref78; Tier 2 (Medium-High)]	Advanced modified NASICON systems have reported approximately 148–165 mAh/g.[ref119; Tier 1 (High)][ref120; Tier 1 (High)]	Values above baseline NVP arise from different compositions and additional redox couples; they are not plain-NVP capacity.
NVPF and related fluorophosphates	Approximately 115 mAh/g cited for a representative baseline. [ref78; Tier 2 (Medium-High)]	Composition-dependent	Higher voltage, rather than exceptionally high capacity, is a central advantage.
P2/O3 layered oxides	No single theoretical value applies across NaxTMO2 compositions	Selected Fe/Mn P2 materials report initial discharge up to 243 mAh/g.[ref78; Tier 2 (Medium-High)]	Initial half-cell capacity may include high-voltage processes that compromise retention.
Hard carbon	No single crystallographic	Typically 250–350 mAh/g; about 300 mAh/g is a	Density, ICE and plateau fraction are as important

Material	Theoretical or reference capacity	Demonstrated or practical capacity	Interpretation
	theoretical limit suitable for all hard carbons	common benchmark. [ref207; Tier 1 (High)][ref10; Tier 3 (Medium)]	as maximum gravimetric capacity.
Advanced doped hard carbon	Not a universal theoretical class	Up to approximately 520 mAh/g reported in selected research.[ref10; Tier 3 (Medium)]	Laboratory result, not a standard commercial assumption.
Tin alloy anode	Approximately 847 mAh/g theoretical for a cited sodium–tin system.[ref287; Tier 3 (Medium)]	Practical retention constrained by expansion	Approximately 300–400% volume expansion can cause mechanical failure. [ref149; Tier 1 (High)]

Specific capacity alone is not cell energy. Average voltage, first-cycle efficiency, electrode balancing, inactive materials, electrolyte quantity and achievable coating density all affect Wh/kg and Wh/L. A low-density cathode or porous hard carbon can provide good gravimetric performance while imposing a volumetric penalty.

Emerging–manufacturer profiles

HiNa Battery

HiNa has one of the clearest non-CATL vehicle records in the supplied evidence. JAC’s Yiwei brand began delivering a sodium–ion passenger vehicle in January 2024, described as the first volume-produced sodium–ion EV.[ref123; Tier 3 (Medium)][ref127; Tier 3 (Medium)] Reported specifications include a 23.2 kWh pack, 230 km CLTC range, 32140 cylindrical cells, more than 140 Wh/kg and 10–80% charging in approximately 20 minutes.[ref122; Tier 3 (Medium)][ref123; Tier 3 (Medium)]

The vehicle’s chemistry was reported as a copper-containing layered–oxide cathode paired with hard carbon.[ref122; Tier 3 (Medium)] Peer-reviewed post-mortem work identified $\text{NaCu}_{1/9}\text{Ni}_{2/9}\text{Fe}_{1/3}\text{Mn}_{1/3}\text{O}_2$ in a HiNa cell, providing stronger evidence for a Cu/Ni/Fe/Mn layered–oxide route.[ref91; Tier 1 (High)] Claims that this vehicle used Prussian white conflict with the more specific layered–oxide evidence and should not be relied upon.

HiNa was also associated with the 100 MW/200 MWh Datang Hubei sodium–ion storage project. [ref92; Tier 2 (Medium–High)] The source supports project deployment context, though cell supplier shares and commissioned operating data should be separately verified for investment-grade analysis.

In March 2025 HiNa announced the Haixing commercial-vehicle family, including K150, K210, K280 and K350 products.[ref121; Tier 3 (Medium)] Reported claims include more than 165 Wh/kg, 20–25

minute charging, more than 8,000 cycles and operation from -40°C to +45°C.[ref126; Tier 2 (Medium-High)] Those are product claims rather than independently validated fleet-life results.

SVOLT

The supplied Reuters evidence contains no specific sodium-ion product, chemistry, production line, customer or deployment information for SVOLT.[ref57; Tier 2 (Medium-High)] SVOLT should therefore be recorded as **not substantiated in this evidence set**, rather than assigned a chemistry or production status from general market lists.

AMTE Power

Reuters identified AMTE Power as a UK public company founded in 2013 and active in sodium-ion development, with investors including Renaissance Capital Partners and Mitsubishi Materials.[ref195; Tier 2 (Medium-High)] The supplied evidence does not establish a current commercial cell chemistry, operating production capacity, customer delivery volume or post-restructuring status as of the cutoff. Its profile is consequently limited to **documented developer**, not verified scaled manufacturer.

Northvolt

Northvolt publicly developed a sodium-ion concept directed toward stationary and industrial storage and associated with Prussian-blue chemistry in industry reporting.[ref67; Tier 2 (Medium-High)] However, the Reuters material supplied for this investigation provides no detailed operational update, production volume or customer deployment.[ref196; Tier 2 (Medium-High)] Northvolt should therefore be classified as a **publicly announced sodium-ion developer**, with commercial status unresolved in the present evidence.

Faradion and Reliance Industries

Reliance Industries acquired UK sodium-ion developer Faradion around December 2021 for approximately £100 million, reported as roughly \$117–130 million depending on source timing and conversion.[ref196; Tier 2 (Medium-High)][ref199; Tier 2 (Medium-High)][ref202; Tier 2 (Medium-High)] Faradion is associated with layered-oxide cathodes and hard-carbon anodes and has targeted mobility, telecommunications backup and stationary storage.[ref165; Tier 3 (Medium)]

The acquisition is a verified corporate event; it is not evidence of a particular GWh output or field deployment. The supplied sources do not provide an audited 2026 production volume, named high-volume vehicle customer or current cell specification. Faradion/Reliance should be viewed as a strategically backed technology platform whose scale must be verified separately.

Natron Energy

Natron specialized in Prussian-blue-analogue chemistry and emphasized high-power, fast-charge and long-cycle applications such as data-center UPS and industrial backup.[ref165; Tier 3 (Medium)] In August 2024 it announced a proposed \$1.4 billion North Carolina facility designed for

24 GW of annual battery output, compared with an existing 600 MW site in Holland, Michigan. [ref201; Tier 2 (Medium-High)] The 24 GW figure was planned capacity, not operating output.

The later IEA assessment states that Natron shut down and uses this as evidence of the difficulty of scaling sodium-ion production outside China.[ref133; Tier 1 (High)] At the October 2026 cutoff, the higher-authority and more recent IEA status should supersede earlier expansion announcements. Natron therefore illustrates the difference between a funded factory plan and durable commercial production.

Sodium-ion versus lithium-ion

Sodium-ion is not one chemistry, and neither is lithium-ion. Comparisons must identify whether the lithium benchmark is LFP or NMC and whether the sodium cell uses PBA, layered oxide or a polyanionic cathode.

Attribute	Sodium-ion	Lithium-ion	Decision implication
Gravimetric cell energy	Broad commercial and prototype ranges are reported around 75–175 Wh/kg; CATL Naxtra claims 175 Wh/kg. [ref304; Tier 1 (High)][ref11; Tier 1 (High)]	IEA comparison: approximately 205 Wh/kg for LFP and 255 Wh/kg for NMC.[ref133; Tier 1 (High)]	Sodium can approach lower-energy lithium products but remains disadvantaged against high-energy NMC.
Volumetric cell energy	Generic range approximately 250–375 Wh/L; often 17–49% below LFP depending on chemistry and format.[ref10; Tier 3 (Medium)][ref299; Tier 1 (High)]	No single lithium-ion Wh/L range is established in the supplied evidence	Volumetric deficit can be more important than gravimetric deficit in vehicles and containerized storage.
Raw materials	Can avoid lithium and cobalt; some chemistries also avoid nickel. Uses sodium salts and can use aluminum on both electrodes.[ref11; Tier 1 (High)] [ref301; Tier 1 (High)]	LFP avoids nickel and cobalt but still uses lithium; NMC uses lithium, nickel, manganese and cobalt.[ref133; Tier 1 (High)]	Sodium diversifies lithium exposure, but layered sodium oxides may still depend on nickel and manganese.
Current collectors	Aluminum can be used on both electrodes.[ref258; Tier 1 (High)]	Aluminum cathode collector and copper anode collector are conventional	Sodium has a structural cost and mass advantage at the collector level.
Cell cost	IEA reports potential production cost up to about 30% below LFP, while noting scale and lithium-price	LFP benefits from highly mature, optimized production.[ref133; Tier 1 (High)]	Sodium's theoretical BOM advantage does not guarantee

Attribute	Sodium-ion	Lithium-ion	Decision implication
	dependence.[ref324; Tier 1 (High)][ref133; Tier 1 (High)]		immediate market-price advantage.
Cycle life	Product-specific: CATL claims more than 10,000 cycles for Naxtra; other commercial targets vary widely.[ref11; Tier 1 (High)]	Chemistry- and use-dependent; no universal comparative figure is supported here	Compare at equal depth of discharge, temperature, C-rate and end-of-life threshold.
Low-temperature performance	CATL claims 90% usable power at -40°C and operation to -50°C for its vehicle system. [ref11; Tier 1 (High)][ref14; Tier 1 (High)]	CATL reports substantially lower equivalent LFP discharge power at -30°C in its own comparison. [ref14; Tier 1 (High)]	Cold-climate mobility and backup power are credible sodium niches, subject to independent validation.
High-temperature operation	Naxtra claims operation to +70°C.[ref11; Tier 1 (High)]	Chemistry-specific; no matching general range supplied	Thermal-system savings cannot be assumed without pack tests.
Safety	Some sodium cathodes are structurally stable, and CATL reports no smoke or fire in cited abuse tests.[ref14; Tier 1 (High)]	LFP is itself a safety-oriented lithium chemistry	“Sodium is inherently safe” is too broad; electrolyte, SOC, format and propagation controls remain decisive.
Zero-volt transport	Aluminum negative collectors permit deep discharge without copper dissolution, but prolonged 0 V storage can damage interfaces.[ref98; Tier 1 (High)]	Copper dissolution can be a concern in overdischarged conventional cells	Sodium may simplify logistics, but storage protocol still matters.
Manufacturing	Many processes can use lithium-ion equipment.[ref20; Tier 1 (High)]	Mature global ecosystem	Compatibility accelerates scaling but does not eliminate sodium-specific yield and moisture challenges.
Supply-chain maturity	Hard carbon and sodium cathode supply chains remain less mature; China holds nearly all current capacity and is projected above 95% through 2030.[ref133; Tier 1 (High)]	Established global graphite, cathode and cell ecosystem	Near-term sodium procurement may remain geographically concentrated despite abundant raw sodium.

Cost should be assessed at system level

The use of sodium salts, iron/manganese-rich cathodes and dual aluminum collectors creates a credible low-cost BOM pathway.[ref301; Tier 1 (High)] Yet cost per kWh also depends on energy density, yield, hard-carbon price, electrolyte quantity, formation time, cell housing, thermal controls and factory utilization. Early sodium-ion production can therefore cost more than its mature-state materials logic suggests.

The IEA concludes that sodium-ion may be up to 30% cheaper to produce than LFP, but also notes that broader competitiveness depends on lithium prices and further energy-density gains.[ref133; Tier 1 (High)][ref324; Tier 1 (High)] Sodium-ion should be evaluated as a hedge against lithium price and concentration risk, not as a guaranteed fixed discount.

Safety claims require chemistry-specific qualification

Prussian-white and polyanionic cathodes can offer strong structural and thermal characteristics, while CATL states that Naxtra eliminates combustion-supporting factors at the material level.[ref11; Tier 1 (High)] Nevertheless, conventional sodium-ion cells still use organic liquid electrolytes unless explicitly described otherwise. Thermal behavior therefore depends on electrolyte quantity, electrode state, cell format and propagation design.

Passing crush, drilling or sawing tests is material evidence for a named design.[ref14; Tier 1 (High)] It does not prove that every sodium-ion chemistry is safer than every LFP design. A valid safety comparison requires common SOC, cell capacity, triggering method, calorimetry and pack-level propagation criteria.

Technical and commercial implications

CATL's portfolio is complementary rather than substitutional

CATL's portfolio separates roles:

- **Shenxing/Shenxing Plus:** LFP fast-charging and range-oriented lithium-ion technology.[ref65; Tier 1 (High)]
- **Naxtra:** sodium-ion technology emphasizing resource resilience, low-temperature operation, safety and selected long-life applications.[ref11; Tier 1 (High)]
- **AB pack:** an early mixed-cell concept combining sodium-ion and lithium-ion cells.[ref20; Tier 1 (High)]
- **Freevoy:** a later dual-power architecture capable of several chemistry combinations, not a synonym for sodium-ion.[ref11; Tier 1 (High)]

This structure reduces the need for one chemistry to satisfy every duty cycle. Sodium-ion can support cold-start power or cost-sensitive energy, while lithium-based zones provide greater energy density. The trade-off is increased BMS, thermal and state-estimation complexity because different cell populations have different voltage curves, resistance, aging and temperature response.

Cathode selection determines market fit

Prussian white is compelling where inexpensive materials, rapid diffusion and high power justify strict moisture and defect control.[ref66; Tier 1 (High)] Layered oxides are attractive for vehicle energy density but require management of phase transitions, air sensitivity and metal-specific supply exposure.[ref82; Tier 1 (High)][ref84; Tier 1 (High)] NASICON and related polyanions favor long-life or high-power applications but can carry conductivity and vanadium-cost penalties.[ref115; Tier 1 (High)][ref119; Tier 1 (High)]

A manufacturer's statement that a cell is "sodium-ion" therefore provides too little information for valuation or procurement. Minimum diligence should identify cathode family, anode type, average voltage, cell energy density, cycle conditions, collector metals, electrolyte class and production status.

Volumetric energy is a critical missing CATL metric

CATL discloses Naxtra's gravimetric energy density but the supplied evidence does not provide a commercial-cell Wh/L value.[ref11; Tier 1 (High)] Hard carbon and open-framework cathodes may have favorable specific capacity yet relatively low tap or electrode density. In passenger vehicles, this can increase pack volume even where cell Wh/kg approaches LFP.

Any claim that Naxtra has "LFP-equivalent energy density" should therefore be read as CATL's gravimetric comparison unless a volumetric basis is separately supplied. Pack-level comparisons also require module structure, cooling, crash protection and cell-to-pack integration.

The industrial bottleneck is reproducibility, not merely laboratory capacity

CATL's reported production challenges—extreme water control, hard-carbon gas generation, aluminum adhesion and self-forming-anode processes—are manufacturing problems rather than deficiencies visible in an ideal coin cell.[ref12; Tier 2 (Medium-High)][ref222; Tier 2 (Medium-High)] Prussian-white water content can impair stability, while hard-carbon surface chemistry controls first-cycle loss and gas.[ref66; Tier 1 (High)][ref394; Tier 1 (High)]

This explains why high reported material capacities cannot be directly converted into factory economics. Yield, swelling, consistency, drying demand, formation inventory and warranty performance determine commercial viability.

Vehicle-user evidence must remain tiered

At the cutoff, the clearest sourced vehicle users are:

- **JAC/Yiwei with HiNa:** reported customer deliveries began in January 2024; chemistry and vehicle specifications are supported primarily by established vehicle media, with layered-oxide chemistry strengthened by peer-reviewed post-mortem evidence.[ref123; Tier 3 (Medium)][ref91; Tier 1 (High)]

- **Changan with CATL:** officially unveiled a Naxtra-equipped mass-production passenger vehicle in February 2026, with market release scheduled for mid-2026.[ref14; Tier 1 (High)] The evidence supplied does not state an audited cumulative delivery number by October 2.
- **Changan Nevo A06:** reported by technology media as the integrated model, but model-level attribution is below the official partnership's verification strength.[ref24; Tier 3 (Medium)]

No other CATL passenger or commercial vehicle should be described as a verified sodium-ion production user solely because it appears in unsourced lists or because an automaker participates in CATL's wider lithium-ion ecosystem.

Stationary storage offers a strong but not risk-free entry point

Stationary systems tolerate lower gravimetric energy better than vehicles and can value cycle throughput, cold operation, current-collector savings and critical-mineral diversification. CATL's TENER Sodium launch and HyperStrong agreement provide strong commercial signals.[ref19; Tier 1 (High)][ref381; Tier 2 (Medium-High)]

However, bankability requires field degradation, warranty terms, round-trip efficiency, augmentation assumptions and safety-certification evidence. A contracted 60 GWh pipeline is not equivalent to 60 GWh of operating assets, and a laboratory cycle count is not equivalent to a warranted project lifetime.

Requirements for the separate Excel workbook

A separate Excel deliverable should contain only values extracted with explicit **unit, basis, chemistry, cell format, test condition, status and source** fields. It should not merge unlike figures into unqualified averages.

Recommended columns are:

Field	Required treatment
Manufacturer/ product	Distinguish first-generation CATL, Naxtra passenger, Naxtra 24 V, TENER Sodium, HiNa vehicle cells and research materials.
Chemistry	Record only the supported resolution: "Prussian white/hard carbon," "layered-oxide composite/hard carbon," or a precise formula where directly identified.
Metric	Capacity, voltage, Wh/kg, Wh/L, cycles, temperature, charge time, pack energy or range.
Value and unit	Preserve original unit; do not convert range claims into point estimates without a documented calculation.
Basis	Active-material, electrode coating, cell, module, pack, vehicle or system.
Test conditions	Temperature, C-rate, SOC window, depth of discharge and end-of-life threshold where available.

Field	Required treatment
Status	Laboratory result, prototype, announcement, certified product, production claim, scheduled delivery, contracted supply or verified customer delivery.
Date	Date of result or announcement.
Source tier	Tier 1–4 under this report’s scheme.
Caveat	Self-reported, independently tested, chemistry unspecified, volumetric value absent, or other limitation.

For example, “175 Wh/kg” should be entered as a CATL Naxtra passenger-cell gravimetric claim, not as a pack value.[ref11; Tier 1 (High)] The “70–85 wt% active material” figures should be entered as selected NVP **electrode-coating formulations**, not whole-cell fractions.[ref263; Tier 1 (High)] [ref264; Tier 1 (High)] The proposed 24 GW Natron facility should be labeled planned capacity announced in 2024 and cross-referenced to the later IEA-reported shutdown, not entered as operating production.[ref201; Tier 2 (Medium-High)][ref133; Tier 1 (High)]

Conclusion

CATL’s sodium-ion development cannot be represented as one unchanged chemistry. Its 2021 first-generation cell was explicitly based on a redesigned Prussian white cathode and porous hard-carbon anode, with up to 160 Wh/kg and an AB pack concept for combining sodium-ion and lithium-ion cells.[ref20; Tier 1 (High)] The prompt’s characterization of that cathode as layered-oxide “NMPA” is therefore unsupported and should be corrected.

Naxtra, launched in April 2025, represents a later industrial platform. CATL claims 175 Wh/kg, more than 10,000 cycles and operation from –40°C to +70°C, while industry reporting identifies a layered-oxide composite cathode and long-cycle hard carbon.[ref11; Tier 1 (High)][ref3; Tier 2 (Medium-High)] The exact oxide formula, electrolyte, separator, electrode recipes, volumetric energy density and full cell BOM remain undisclosed in the supplied evidence.

CATL’s lithium products must remain separate from this account. Shenxing Plus is an LFP battery, not sodium-ion.[ref65; Tier 1 (High)] The 2024 record does not establish sodium chemistry inside Shenxing. Mixed-chemistry integration is instead supported by the 2021 AB pack and later Freevoy architecture.[ref20; Tier 1 (High)][ref11; Tier 1 (High)]

By October 2, 2026, CATL had progressed beyond a laboratory announcement: Naxtra had been launched and certified; a Changan vehicle program had been officially unveiled; TENER Sodium had been announced; and a 60 GWh, three-year HyperStrong contract had been reported.[ref47; Tier 1 (High)][ref14; Tier 1 (High)][ref19; Tier 1 (High)][ref381; Tier 2 (Medium-High)] Nevertheless, scheduled releases and contracted volumes must not be portrayed as completed deployments. Public evidence still lacks an audited cumulative vehicle-delivery count and commissioned Naxtra storage volume at the cutoff.

Across the industry, sodium-ion is diversifying rather than converging. HiNa provides evidence for Cu/Ni/Fe/Mn layered oxides in an urban EV; Prussian-blue systems target high-power and stationary duty; NASICON and polyanionic research prioritize structural life; and hard carbon remains the dominant practical anode despite first-cycle and process-control challenges.[ref91; Tier 1 (High)][ref115; Tier 1 (High)][ref394; Tier 1 (High)]

The commercial proposition is therefore conditional. Sodium-ion can reduce exposure to lithium, cobalt and copper, perform strongly in cold conditions and leverage much of the lithium-ion manufacturing base.[ref20; Tier 1 (High)][ref301; Tier 1 (High)] Its limitations include lower gravimetric and volumetric energy than leading lithium-ion cells, immature hard-carbon and cathode supply chains, uncertain early-production cost, and insufficient long-duration field evidence.[ref133; Tier 1 (High)][ref299; Tier 1 (High)]

The technically defensible conclusion is not that sodium-ion will universally replace lithium-ion. It is that sodium-ion has become a credible complementary platform for cold-climate vehicles, urban mobility, start-stop systems and stationary storage—provided that chemistry, basis, test conditions and deployment status are stated explicitly.

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Tier 1 (High)

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